Fiber Optic Network Design

College of Opt. Sci. & Engr., ZJU

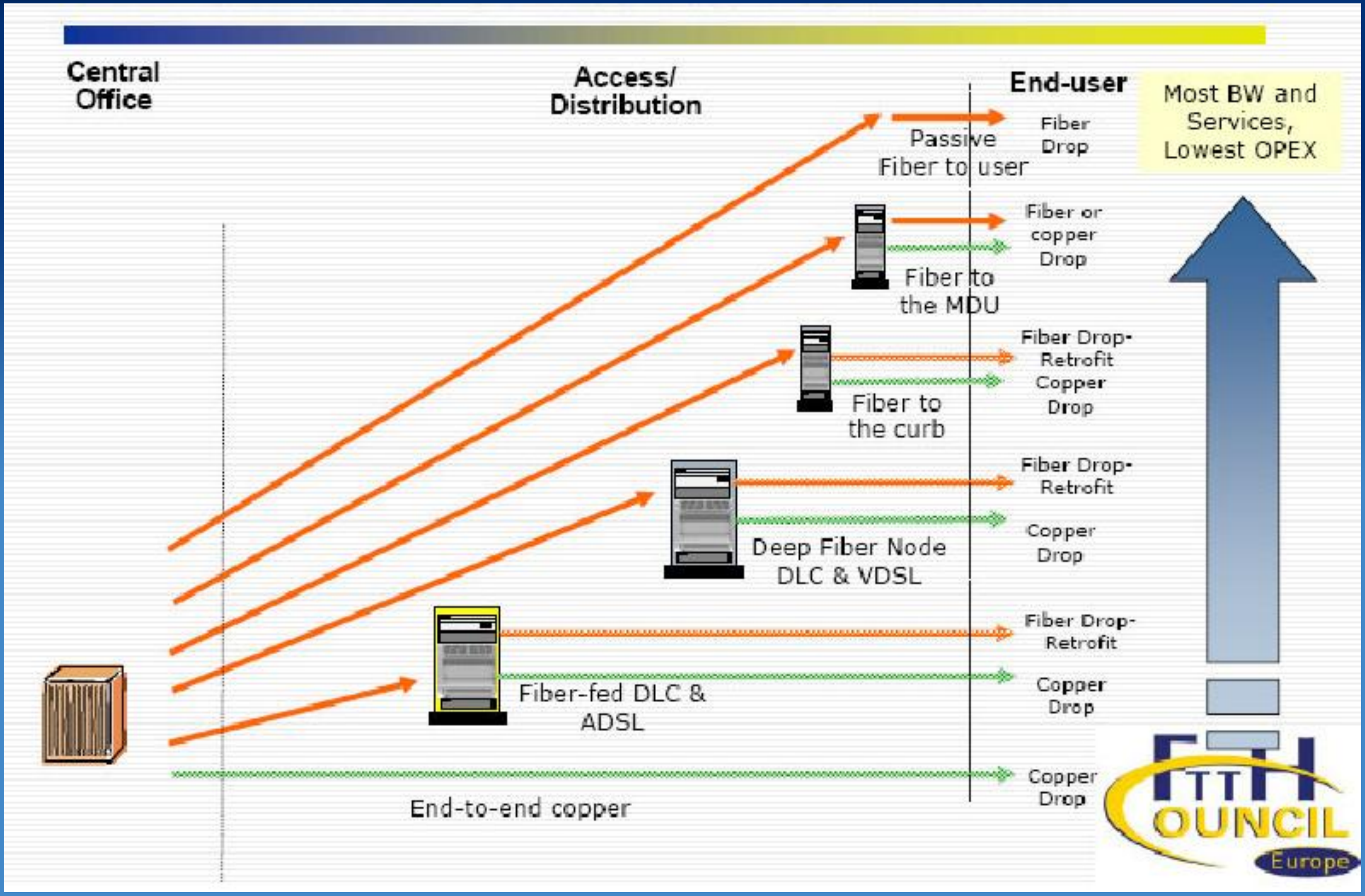
2026

Outline

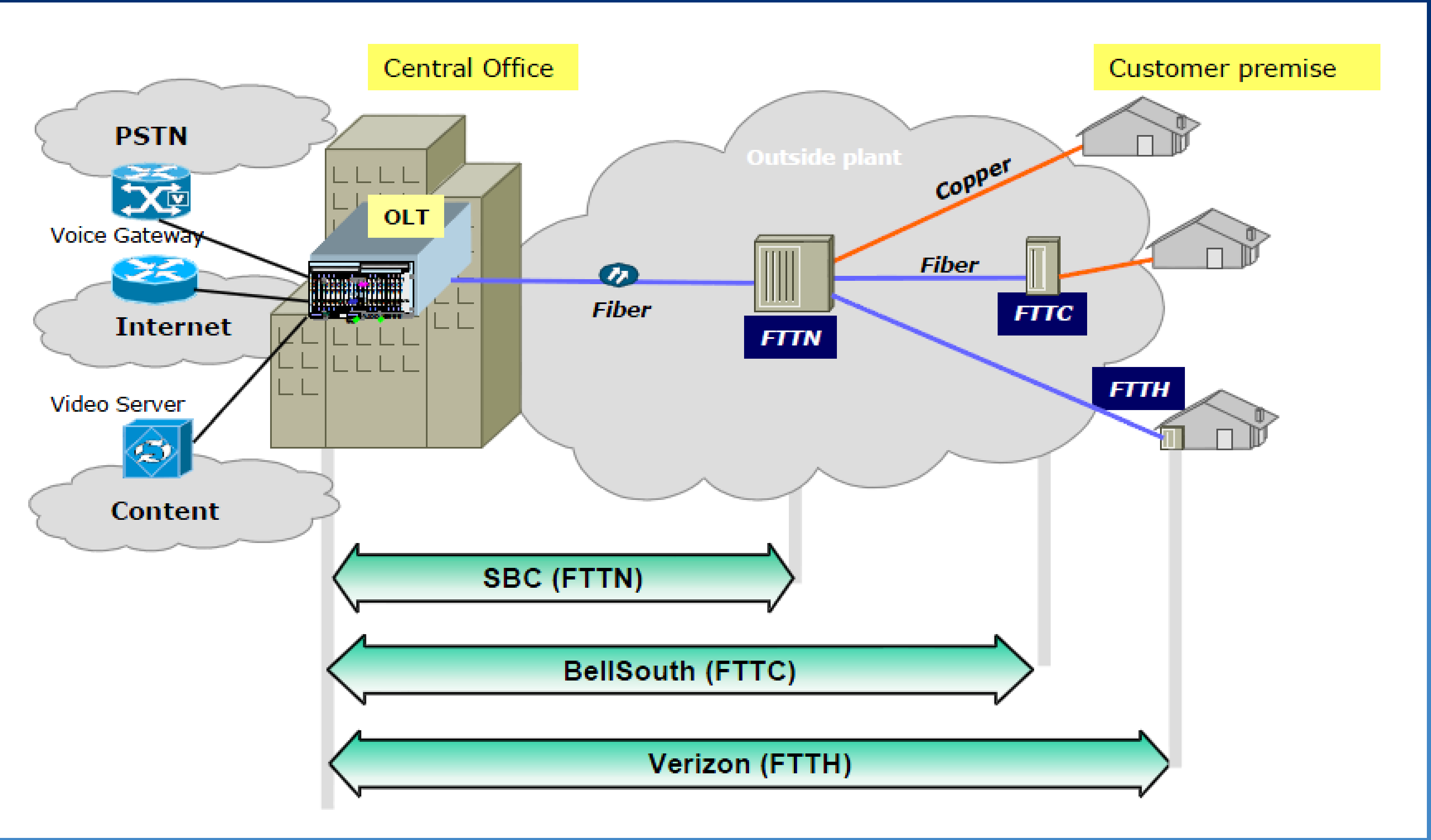
Access Network

- Motivation and Background
- FTTx approaches
    - Point-to-point
    - AON
    - TDM-PON (BPON, EPON,GPON)
    - WDM-PON
- Exercise

Access Network Evolution



Major Approaches in US



Legacy Copper Wire Solutions

Type of Service	Max Downstream Capacity	Max Upstream Capacity	BW*Distance Limit
ADSL	8 Mbps	0.8 Mbps	< 3.6 km
ADSL2+	25 Mbps	1 Mbps	FTTN (< 3 km)
VDSL	52 Mbps	6 Mbps	FTTC (< 300 m)
HFC	3 to 6 Mbps	.125 to .512 Mbps	Depends on sharing

Values are only approximations
Actual rates are typically smaller

Fiber can support > 1,000,000 Mbps

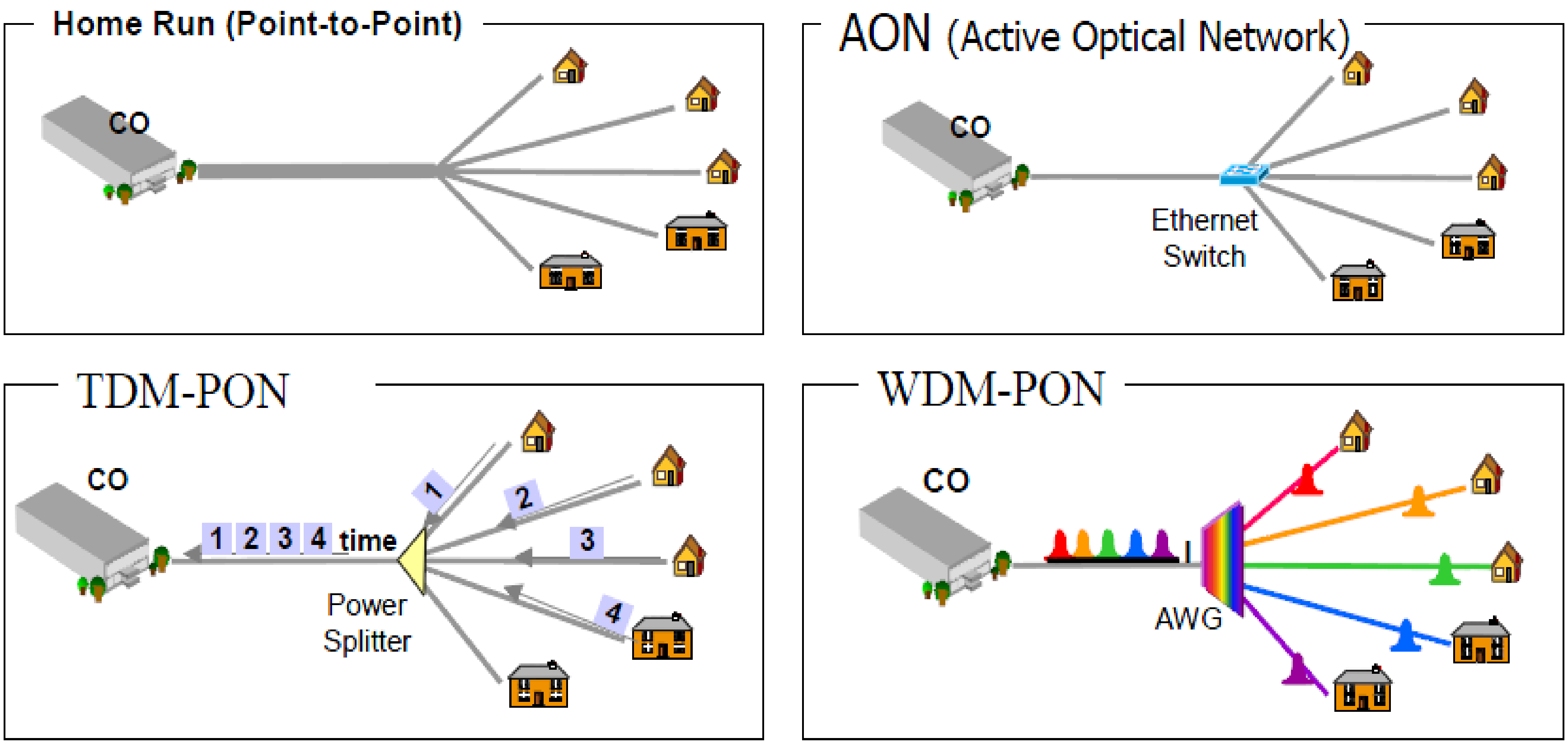
Near Term Applications

HDTV (IPTV, VoD) 1 to 3 ch	30 Mbps
Interactive Video (gaming, EoD, video conf)	10 Mbps
Internet Browsing	5 Mbps
Voice (VoIP)	0.1 Mbps

Minimum requirements for a new network **> 50 Mbps**

Future needs will be much greater (peer-peer, telemedicine, 3D-TV, ….)

FTTx Approaches

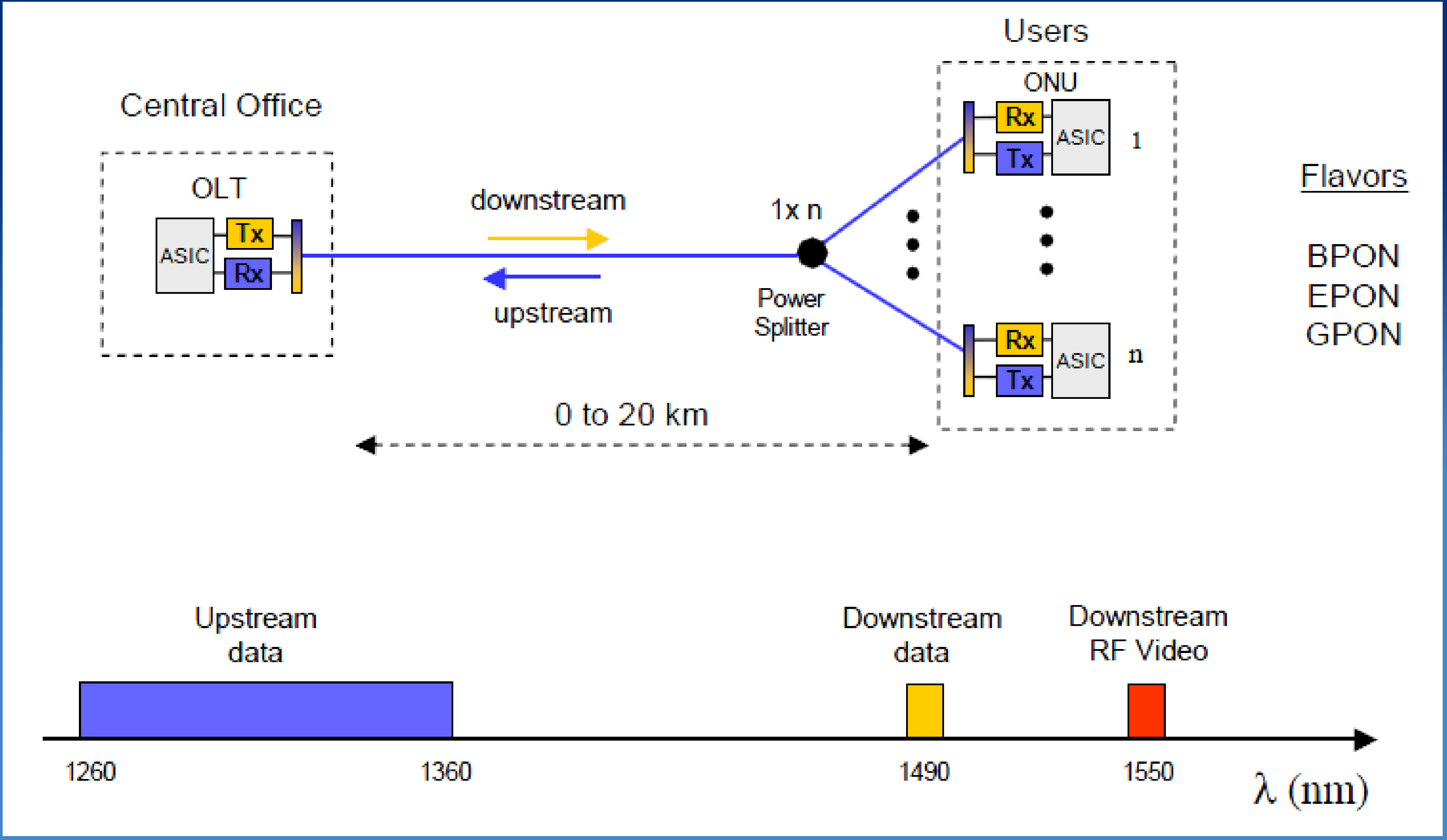


Source: Korea Telecom

Lots of Names and Confusion

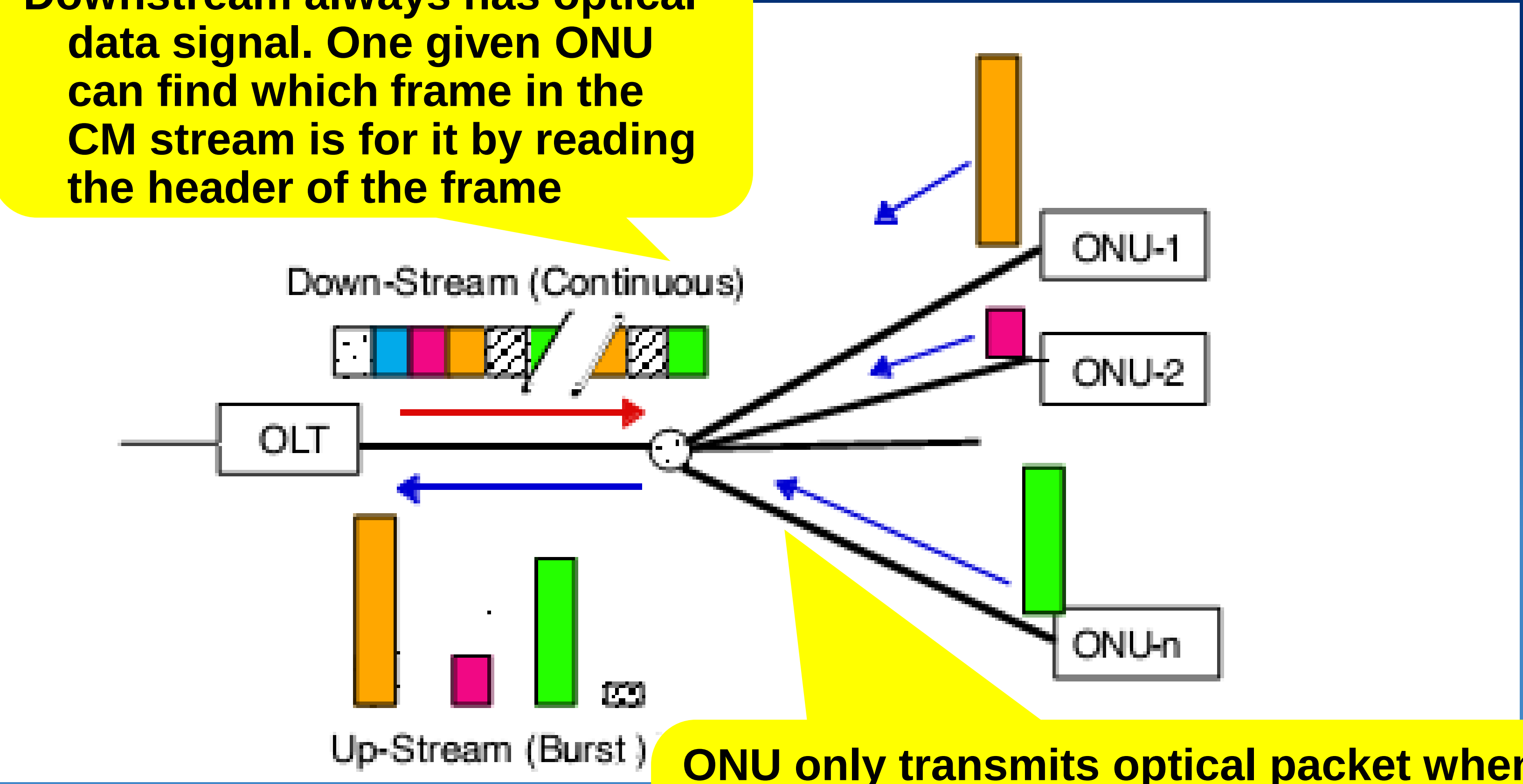
- FTTx = Fiber-to-the-x
  - FTTH - Home
  - FTTC - Curb
  - FTTN - Node or Neighborhood
  - FTTP - Premise
  - FTTB - Building or Business
  - FTTU - User
  - FTTZ - Zone
  - FTTO - Office
  - FTTD - Desk

TDM-PON Basics



Typical PON Configuration and Optical Packets

Downstream always has optical data signal. One given ONU can find which frame in the CM stream is for it by reading the header of the frame

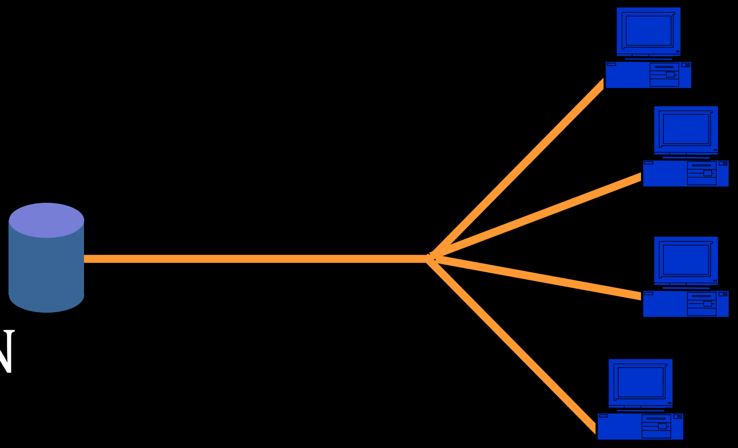


ONU only transmits optical packet when it is allocated a time slot and it needs to transmit, and all the ONUs share the upstream channel in the time division multiple access (TDMA) mode.

Types of PON

many types of PONs have been defined

APON	ATM PON
BPON	Broadband PON
GPON	Gigabit PON
EPON	Ethernet PON
GEPON	Gigabit Ethernet PON
CPON	CDMA PON
WPON	WDM PON



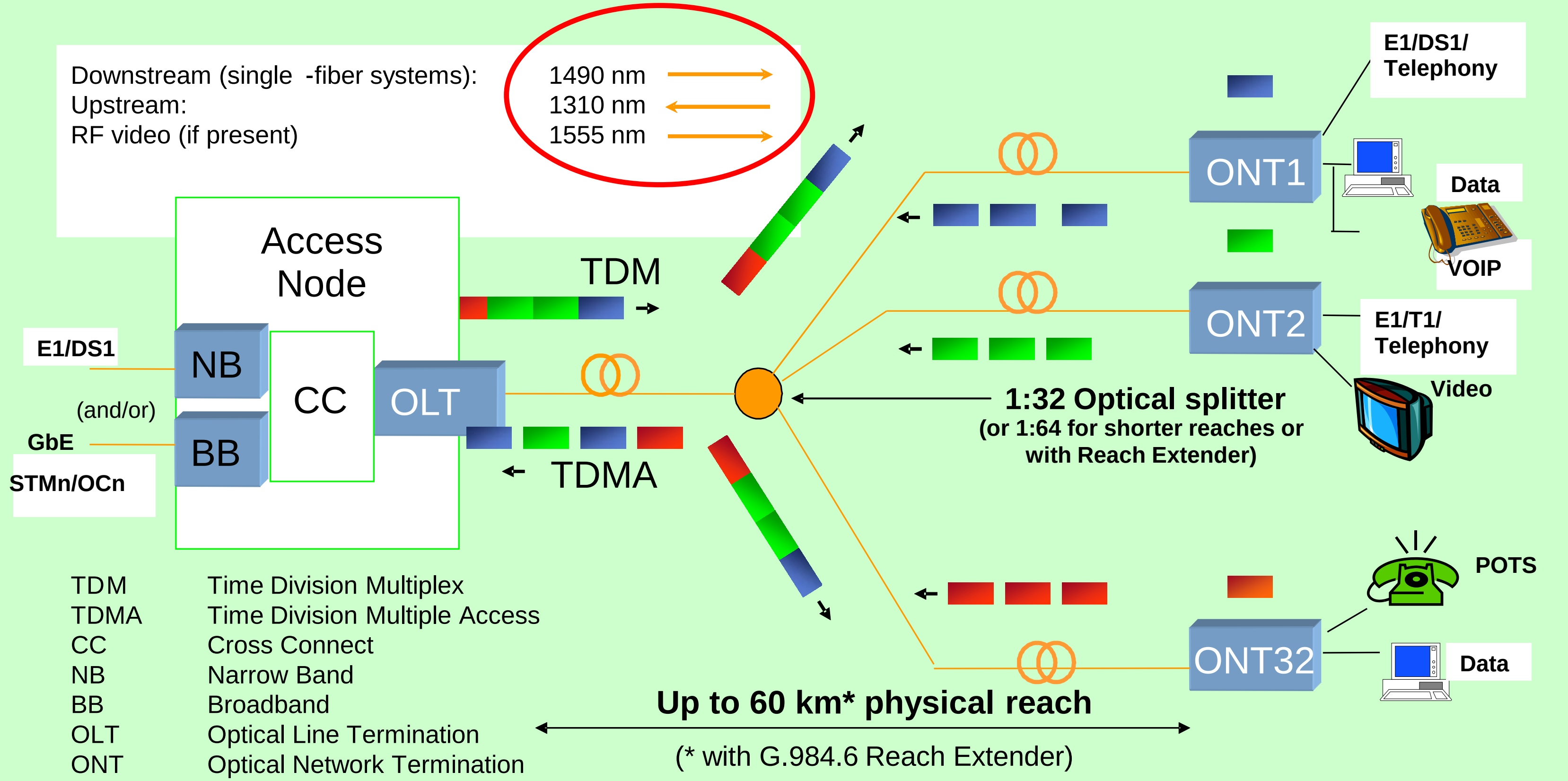
in this course we will focus on GPON and EPON (including GEPON) with a touch of BPON thrown in for the flavor

EPON是怎么实现宽带的?

- EPON接入网的第一级是OLT (optical line terminal)设备，就是局端设备，主要用来对分布的用户端ONU设备进行集中管理，并提供上下行业务控制和QOS (Quality of Service)保障，目前主要通过DBA (Database Administrator)来完成带宽分配；
- 下一级是光分支器OND （optical distribution node）；目前有1:32、1:64的分支，主要是连接终端设备的树形结构；
- 终端设备是ONU （optical network unit），在GPON里面叫ONT,提供用户接入；上下行1G，用户UNI口一般根据ONU种类不同名称不同，大容量的叫MDU，可以提供24口FE或者更多，VOIP/E1业务等。小容量直接放用户家的那种叫SFU，也是FTTH/FTTB的区别；再往下就是用户家的设备了，或者直接PC或者家庭网关。

TDM PON Example

- **Downstream** – TDM transmission with multiple "listeners" (encryption to insure privacy)
- **Upstream** – TDMA transmission with upstream transmissions (bursts) scheduled to prevent overlap



PONs are (in some sense) like HFC systems – shared medium

Optical Splitters

1xN PLC OPTICAL SPLITTER

PRODUCT SPECIFICATIONS[1]

	1X4	1x8	1x16	1x32
	"Premium" Grade			
Maximum Insertion Loss[2] (dB)	7.0	10.3	13.4	16.8
Typical Insertion Loss (dB)	6.6	9.6	12.8	16.0
Uniformity (dB)	< 0.5	< 0.5	< 0.8	< 0.9
	"A" Grade			
Maximum Insertion Loss (dB)	7.4	10.7	14.0	17.2
Typical Insertion Loss (dB)	7.0	10.1	13.5	16.5
Uniformity (dB)	< 0.6	< 0.8	< 1.0	< 1.1
Operating Wavelength (nm)	1250-1650			
Maximum PDL[3] (dB)	< 0.15			< 0.2
Return Loss (dB)	> 55			
Operating Temperature (°C)	-40 to +85			
Storage Temperature (°C)	-40 to +85			
FTTH Tray Package[4] (mm)	40x4x4			55x7x4
Rack[5]	19" 1U			
Fiber Type (standard)	In: SMF-28e Tight Buffer 900um, 1±0.1m Length Out: SMF-28e Bare Ribbon 1±0.1m Length			

1x32 SC/APC Optical Splitter Module

Budget Calculations

$$L_B = |P_S| - |P_O|$$

L_B = Link Budget, P_S = Sensitivity, P_O = Output Power

Example:

GPON 1310nm

Power: 0dBm Single-mode fiber

Sensitivity: -23dBm

Link Budget: 23dB

Typical Range Calculation

$$R_B = \frac{L_B - SIL - 2}{0.35}$$

Assume:

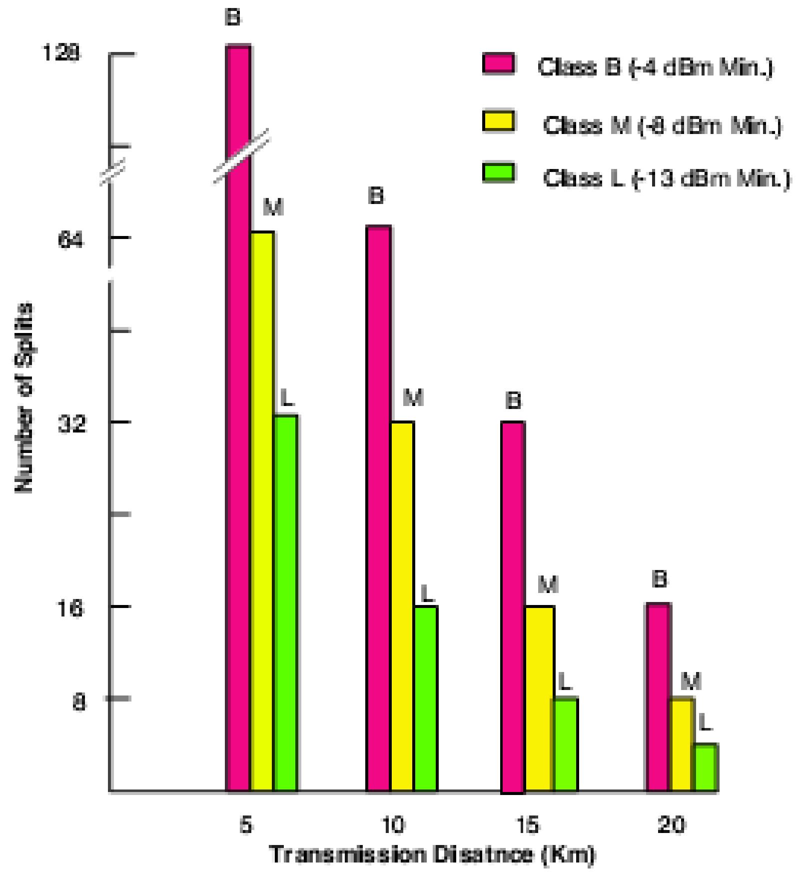
Optical loss = 0.35 dB/km

Connector Loss = 2dB

Splitter Insertion Loss 1X32 = 17dB

Range Budget: ~11km

Relationship between transmission distance and number of splits



The relationship between transmission distance and number of splits, using the transmitter optical output as a parameter, where optical loss in fiber of 0.35 dB/km and insertion/connectors loss of 2 dB were assumed.

PON physical layer

λ Allocations： stage 1- G.983.1

Upstream and downstream directions need about the same bandwidth

US serves N customers, so it needs N times the BW of each customer

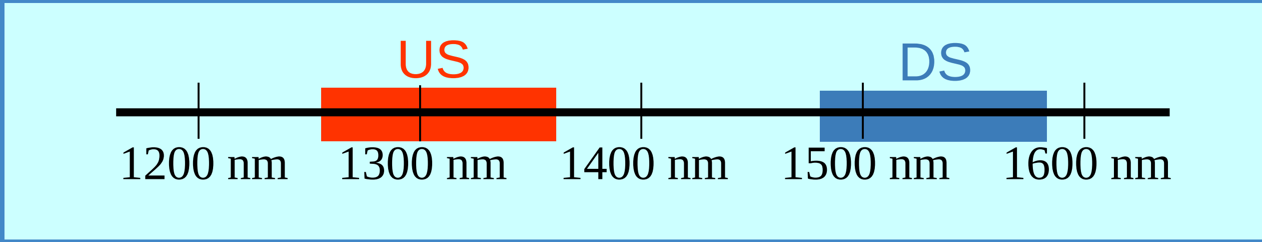
but each customer can only transmit 1/N of the time

In APON and early BPON： 100 nm was needed

Where should these bands be placed for best results?

In the *second* and *third windows* !

- Upstream 1260 - 1360 nm (1310 $\pm$ 50) ***second window***
- Downstream 1480 - 1580 nm (1530 $\pm$ 50) ***third window***



λ Allocations: stage 2 - G.983.3

Afterwards it became clear that there was a need for additional DS bands

Pressing needs were **broadcast video and data**

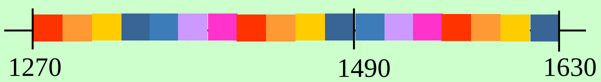
Where could these **new DS bands** be placed ?

At about the same time G.694.2 defined 20 nm CWDM bands

these were made possible because of new inexpensive hardware (uncooled **D**istributed **F**eed**b**ack Lasers)

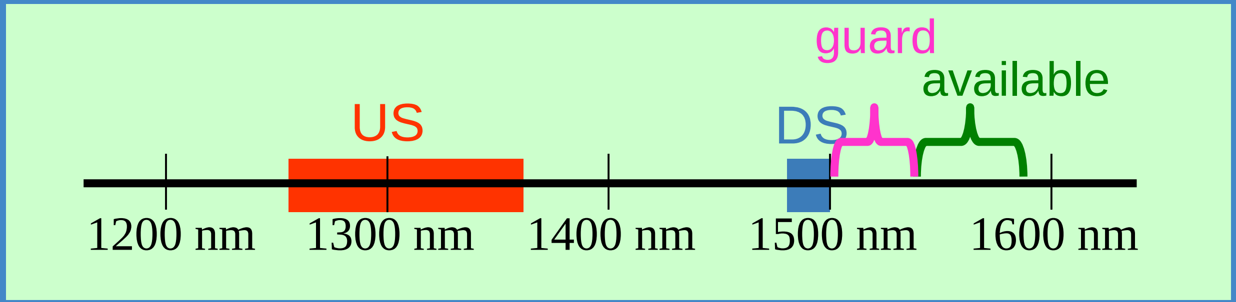
One of the CWDM bands was 1490 ± 10 nm

same bottom λ as the G.983.1 DS

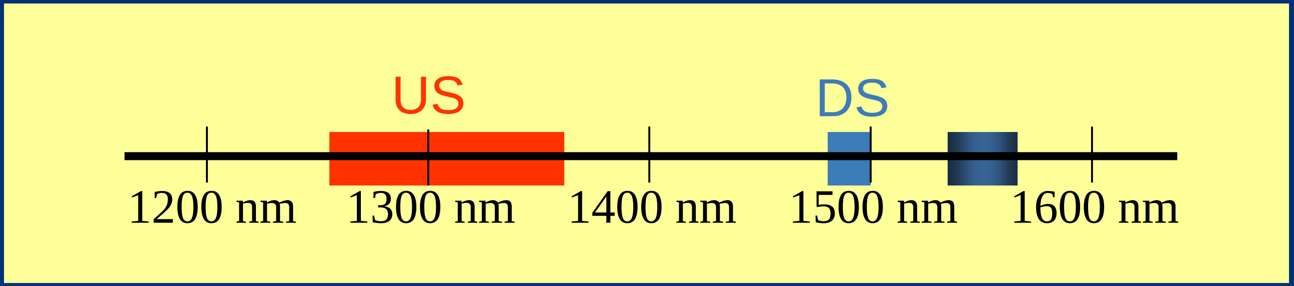


So it was decided to use this band as the G.983.3 DS

and leave the US unchanged



λ allocations - Final



The G.983.3 band-plan was incorporated into GPON
and via liaison activity into EPON
and is now the universally accepted xPON band-plan

- **US 1260-1360 nm (1310 ± 50)**
- **DS 1480-1500 nm (1490 ± 10)**
- enhancement bands:
  - video 1550 - 1560 nm (see ITU-T J.185/J.186)
  - digital 1539-1565 nm

Data rates (for now …)

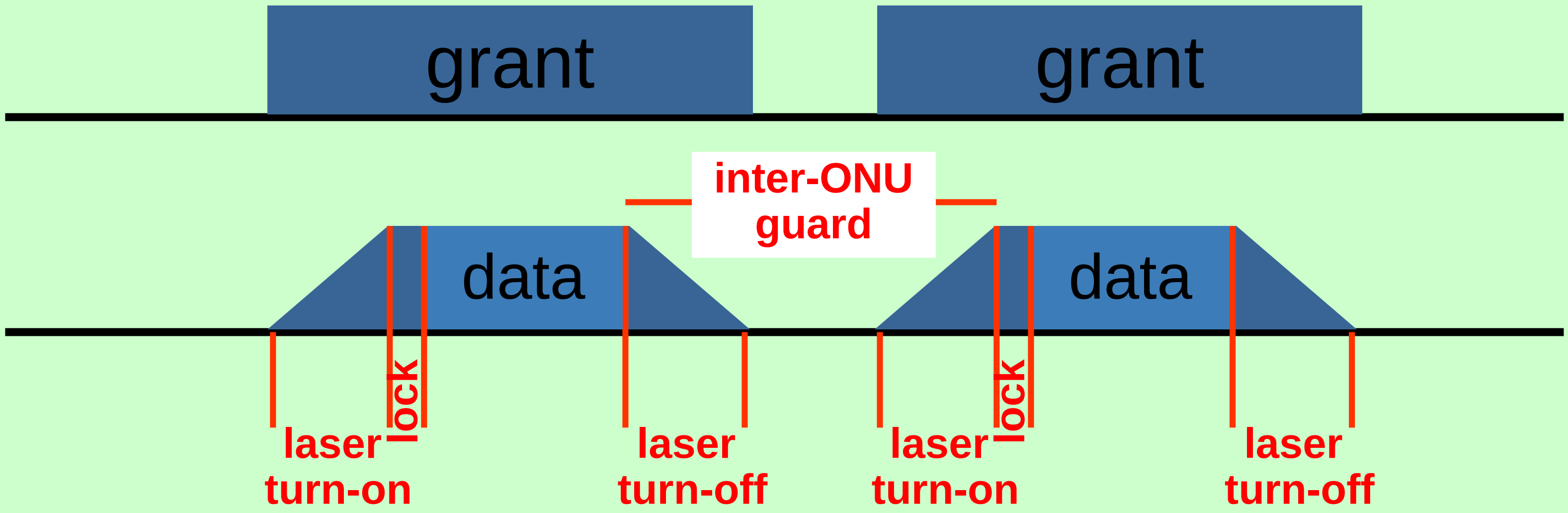
PON	DS (Mbps)	US (Mbps)
BPON	155.52	155.52
	622.08	155.52
Amd 1	622.08	622.08
	1244.16	155.52
Amd 2	1244.16	622.08
	1244.16	155.52
GPON	1244.16	622.08
	1244.16	1244.16
	2488.32	155.52
	2488.32	622.08
	2488.32	1244.16
	2488.32	2488.32
	1250*	1250*
10GEPON†	10312.5*	10312.5*

* only 1G/10G usable due to linecode

† work in progress

Up Stream timing diagram

How does the ONU US transmission appear to the OLT ?



Notes:

GPON - ONU reports turn-on and turn-off times to OLT
ONU preamble length set by OLT

EPON - long lock time as need to Automatic Gain Control and Clock/Data Recovery
long inter-ONU guard due to AGC-reset
Ethernet preamble is part of data

PON User plane

Labels

In an ODN there is one OLT, but many ONUs

ONUs must somehow be labeled for

- OLT to identify the destination ONU
- ONU to identify itself as the source

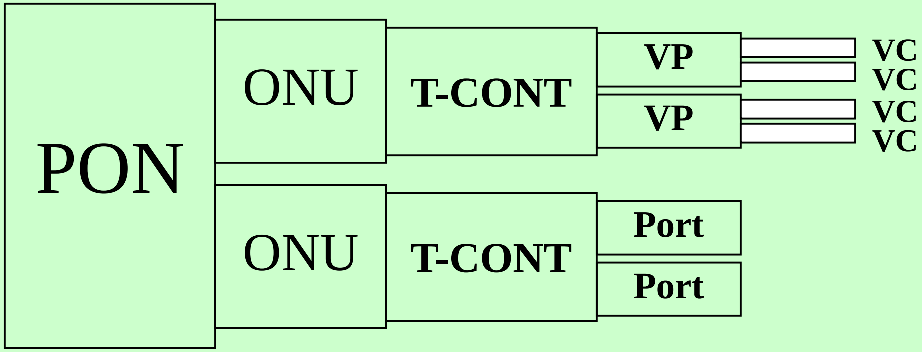
EPON assigns a single label **L**ogical **L**ink **ID** to each ONU **(15b)**

GPON has **several levels of labels**

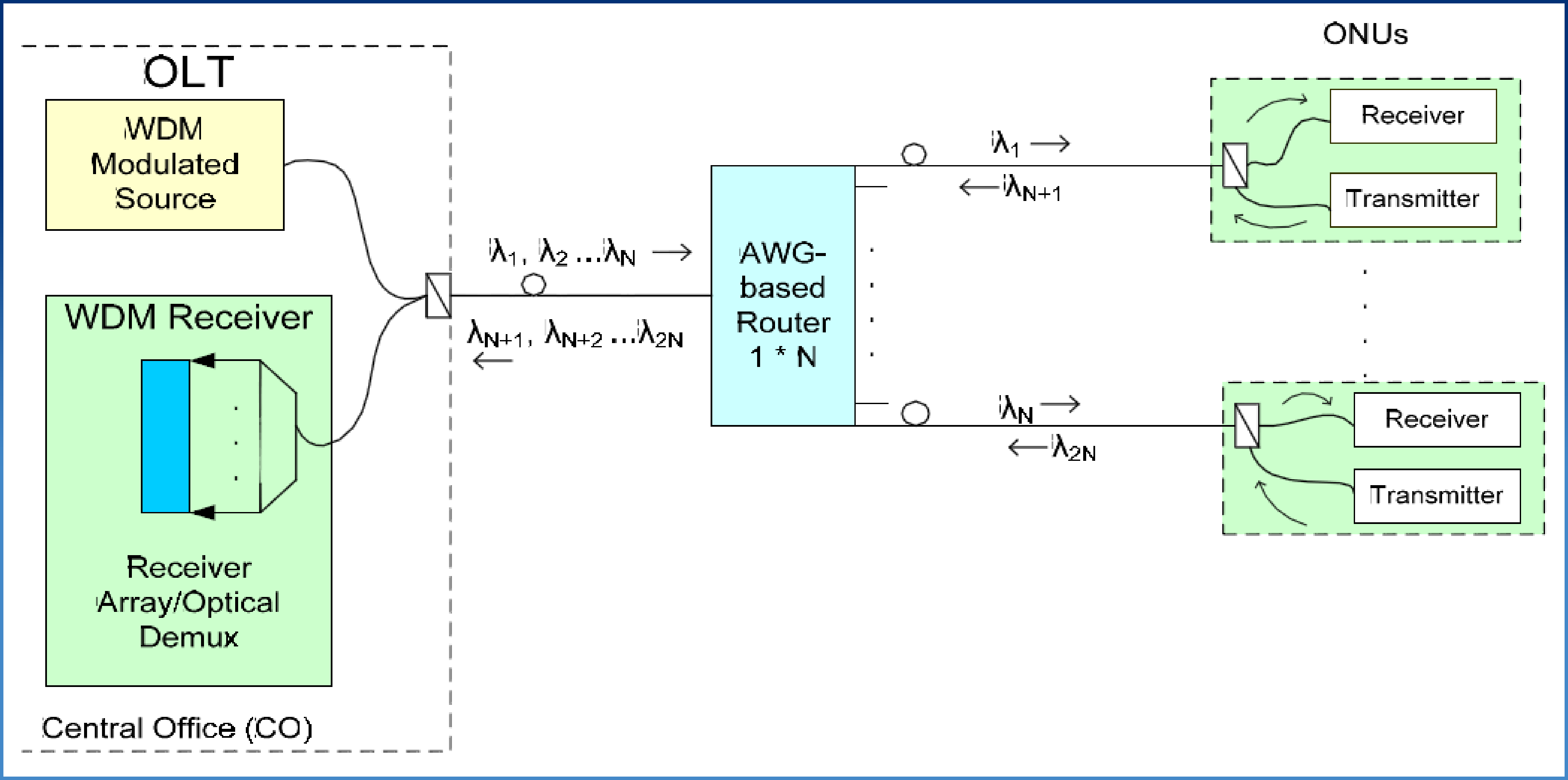
- ONU_ID (1B) **(1B)**
- **T**ransmission-**CONT**ainer (AKA **Alloc_ID**) **(12b)** **(can be >1 T-CONT per ONU)**

  For ATM mode
  - **VPI**
  - **VCI**

  For GEM mode
  - **Port_ID (12b)** **(12b)**



WDM-PON Including Downstream and Upstream



Length v. s. BER

		Length (km)		
		1	8	16
Bit Rate	622 Mbps	5.28E-16	7.63E-15	1.71E-06
	1.2 Gbps	8.39E-07	2.14E-06	4.14E-04
	2.4 Gbps	2.28E-05	2.08E-03	2.06E-02
	10 Gbps	0.017566	1.00E-00	1.00E-00

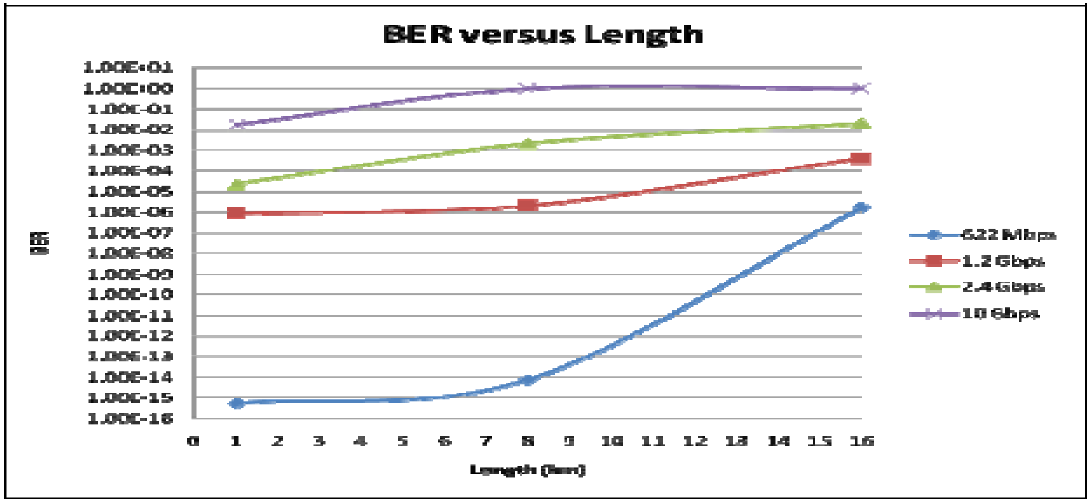
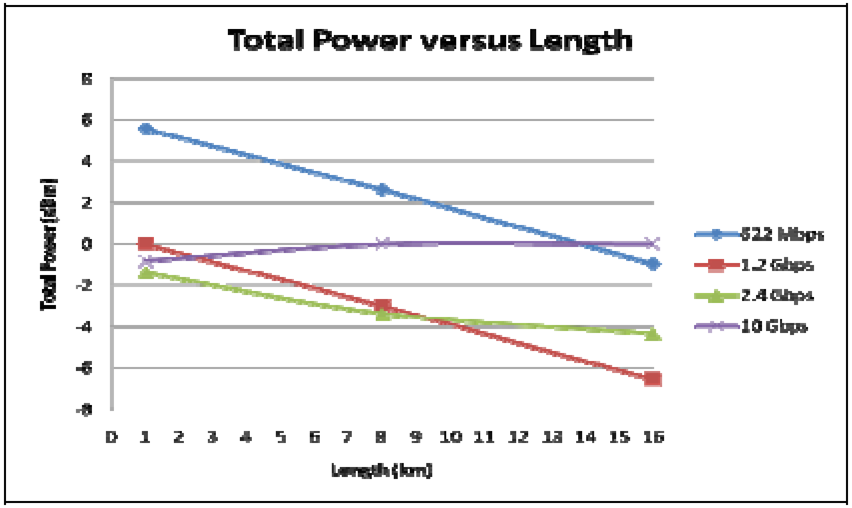


Figure 4 BER vs. length at 5 dBm power transmit

Total Power Received vs. Length

Table 2 Total power received vs. length at 5 dBm power transmit

		Length (km)		
		1	8	16
Bit Rate	622 Mbps	5.571204 dBm	2.600449 dBm	-0.93037 dBm
	1.2 Gbps	0.007866 dBm	-2.97748 dBm	-6.53993 dBm
	2.4 Gbps	-1.3716 dBm	-3.35306 dBm	-4.32667 dBm
	10 Gbps	-0.80807 dBm	0 dBm	0 dBm



Example: 60km GPON with Distributed Raman

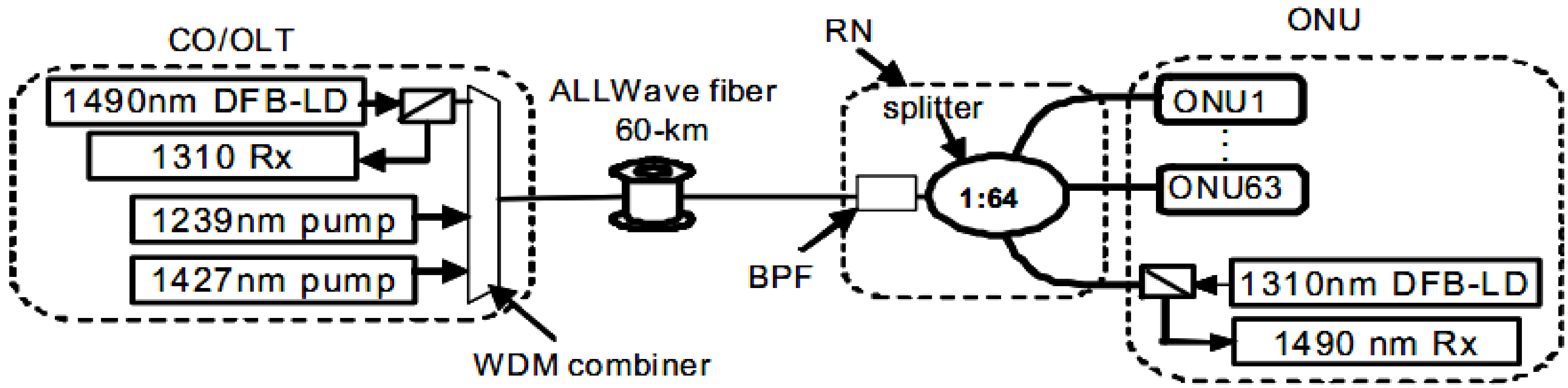


Fig. 1: Experimental set-up for un-powered extender in PON

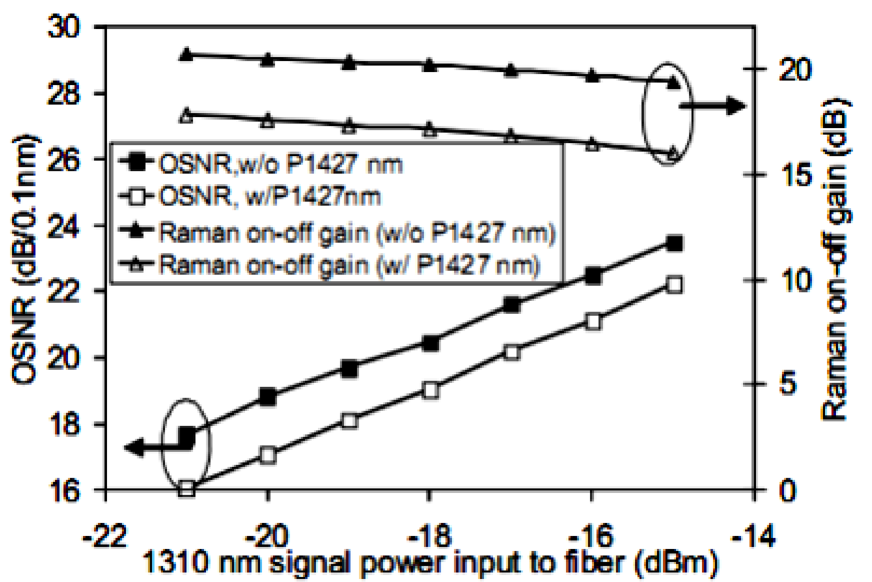


Fig. 3: Raman on-off gains and OSNRs for 1310 signals

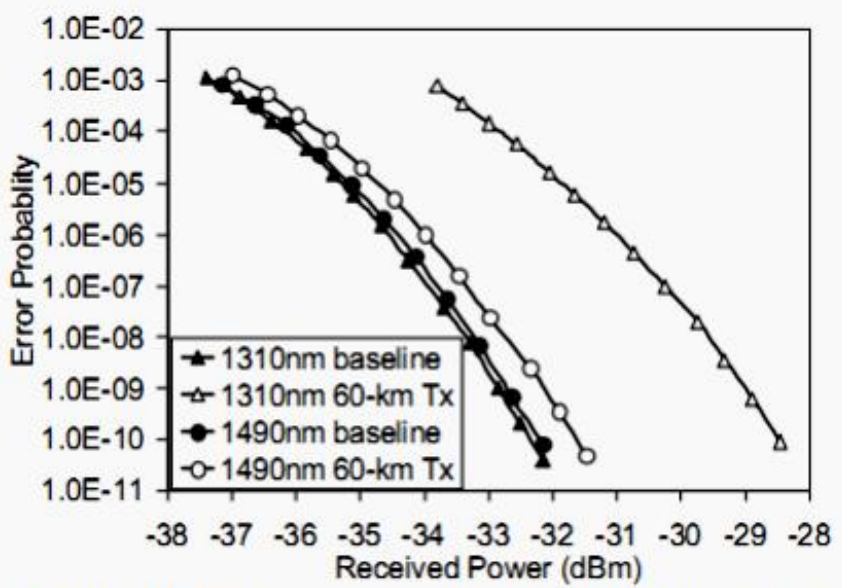
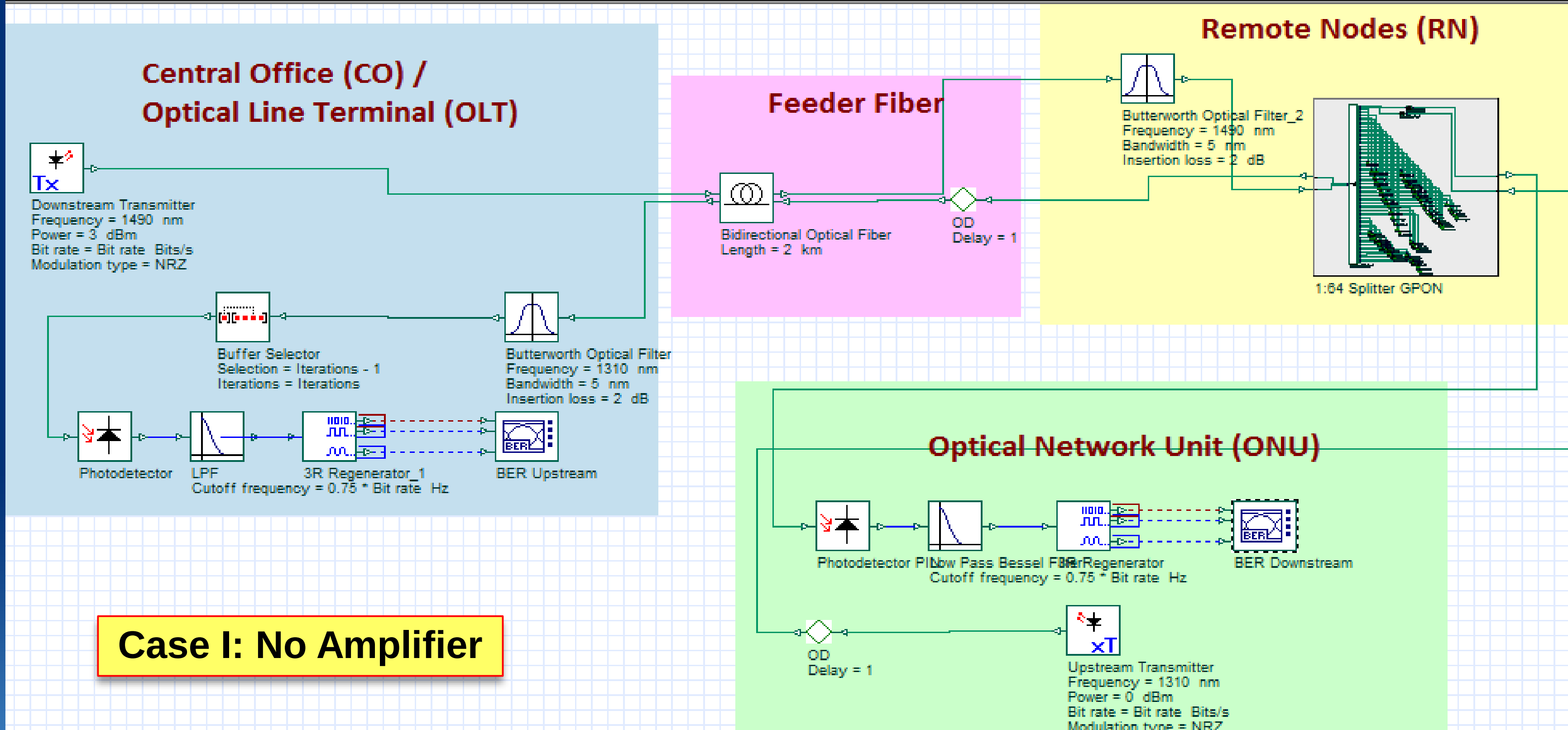
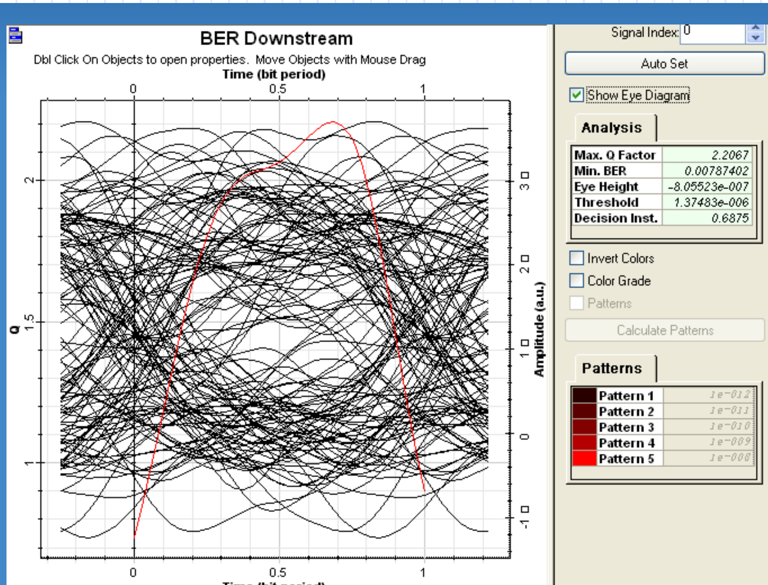
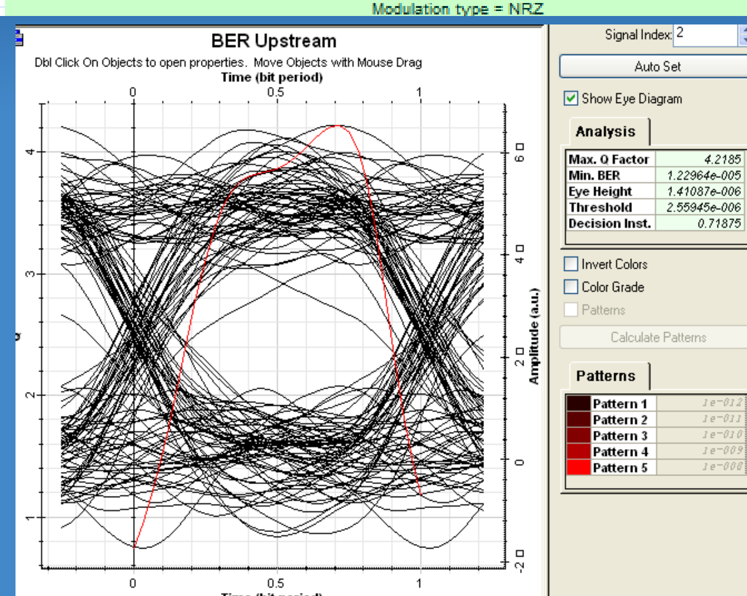


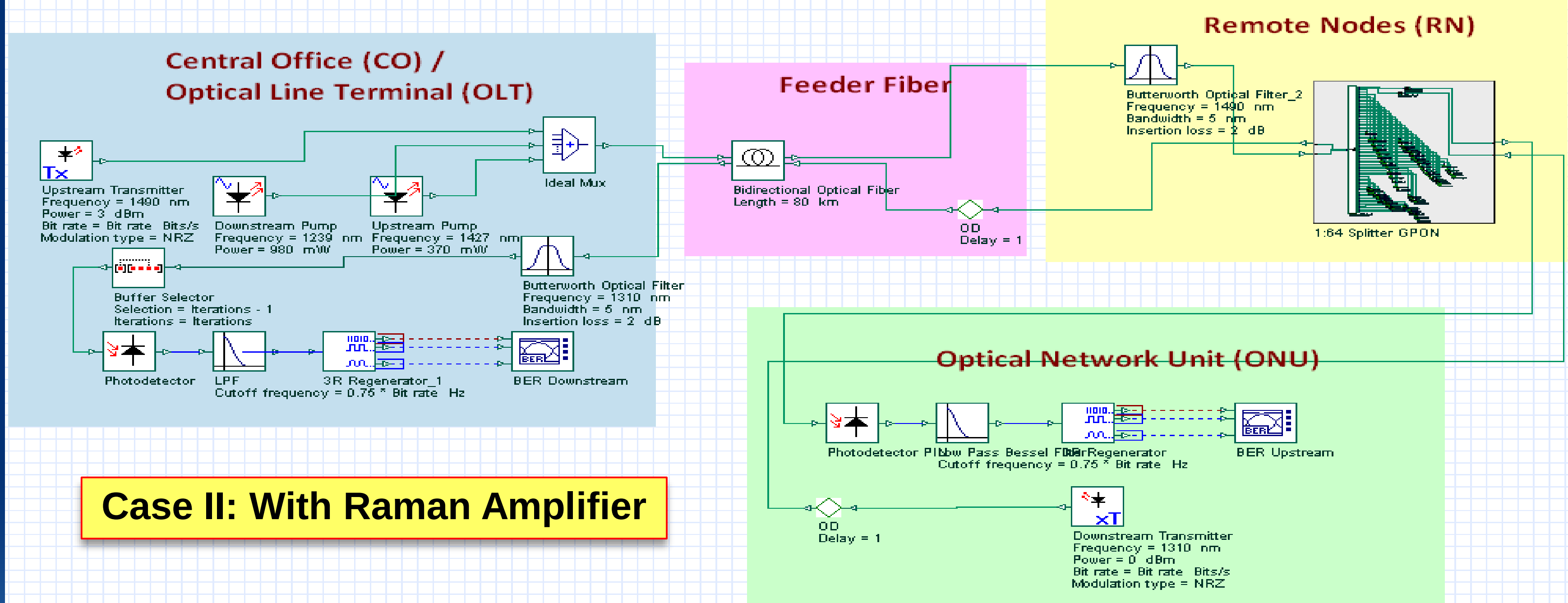
Fig. 4: BER performance of the extended PON systems


Central Office (CO) /
Optical Line Terminal (OLT)
Downstream Transmitter
Frequency = 1490 nm
Power = 3 dBm
Bit rate = Bit rate Bits/s
Modulation type = NRZ
Buffer Selector
Selection = Iterations - 1
Iterations = Iterations
Butterworth Optical Filter
Frequency = 1310 nm
Bandwidth = 5 nm
Insertion loss = 2 dB
Photodetector
LPF
Cutoff frequency = 0.75 * Bit rate Hz
3R Regenerator_1
BER Upstream
Feeder Fiber
Bidirectional Optical Fiber
Length = 2 km
OD
Delay = 1
Remote Nodes (RN)
Butterworth Optical Filter_2
Frequency = 1490 nm
Bandwidth = 5 nm
Insertion loss = 2 dB
1:64 Splitter GPON
Optical Network Unit (ONU)
Photodetector PIN
Low Pass Bessel Filter
3R Regenerator
Cutoff frequency = 0.75 * Bit rate Hz
BER Downstream
OD
Delay = 1
Upstream Transmitter
Frequency = 1310 nm
Power = 0 dBm
Bit rate = Bit rate Bits/s
Modulation type = NRZ
Case I: No Amplifier

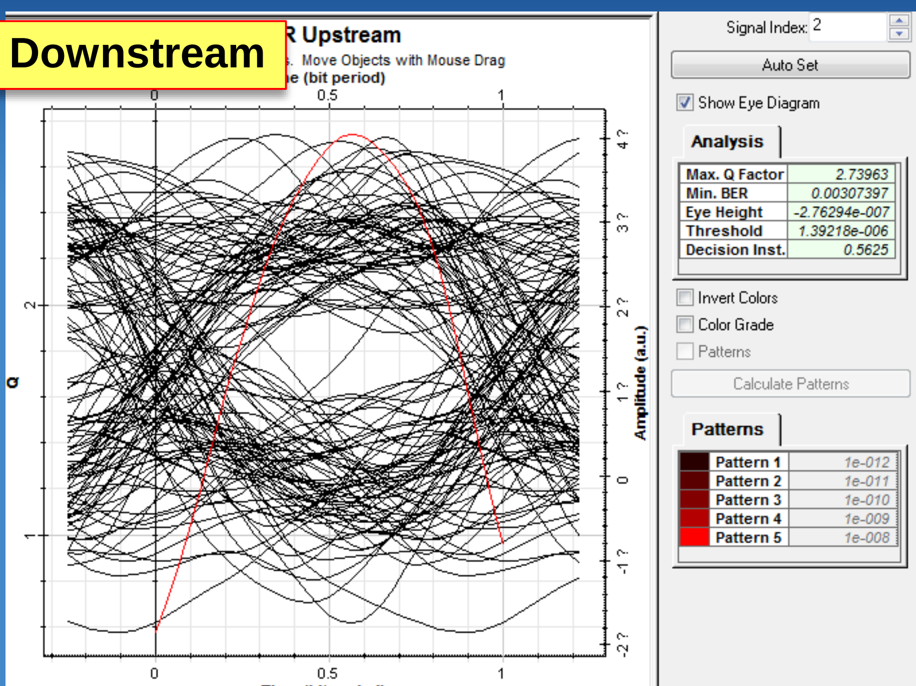

BER Downstream
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Amplitude (a.u.)
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 2.2067
Min. BER 0.00787402
Eye Height -8.05523e-007
Threshold 1.37483e-006
Decision Inst. 0.6875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1
Pattern 2
Pattern 3
Pattern 4
Pattern 5

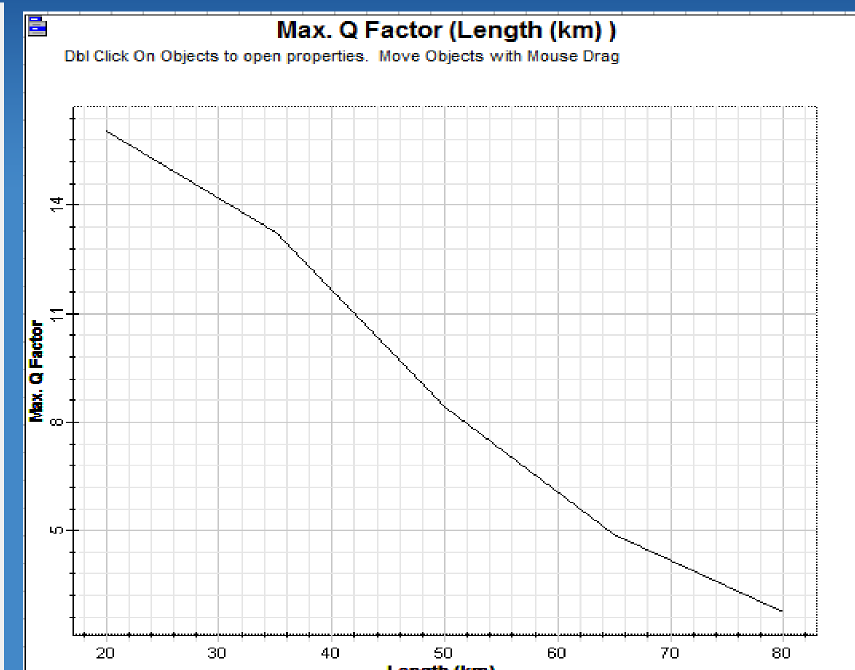

BER Upstream
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Time (bit period)
Amplitude (a.u.)
Signal Index: 2
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 4.2185
Min. BER 1.22964e-005
Eye Height 1.41087e-006
Threshold 2.55945e-006
Decision Inst. 0.71875
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1
Pattern 2
Pattern 3
Pattern 4
Pattern 5

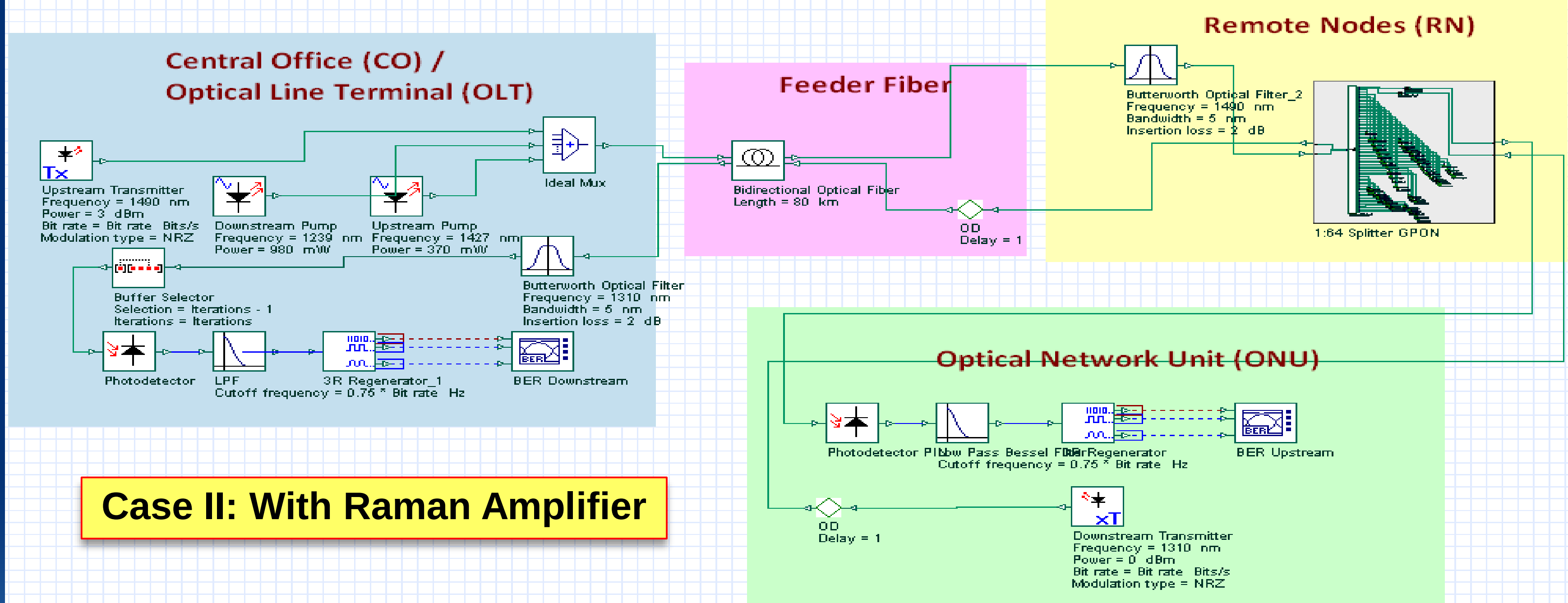
Case II: With Raman Amplifier

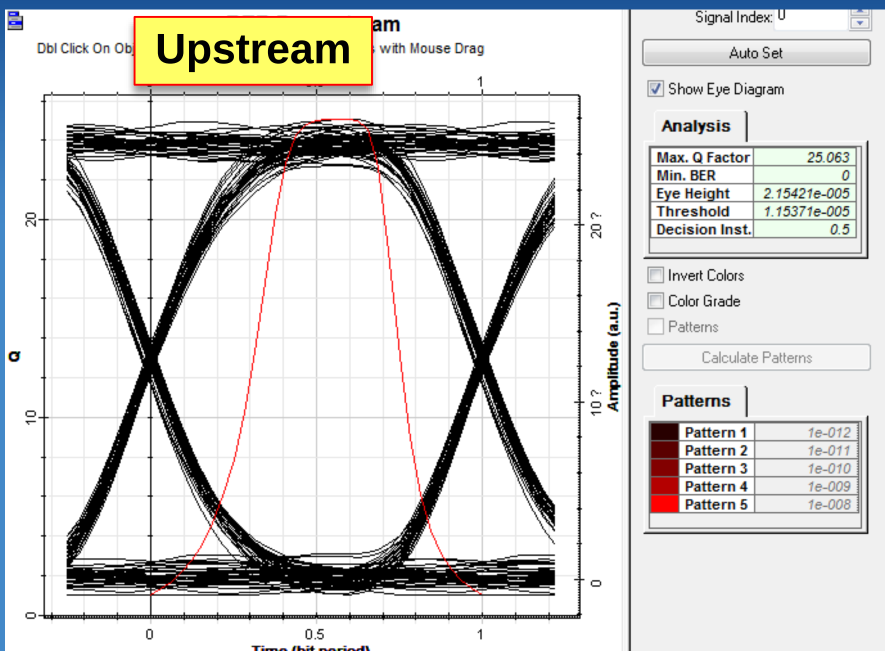


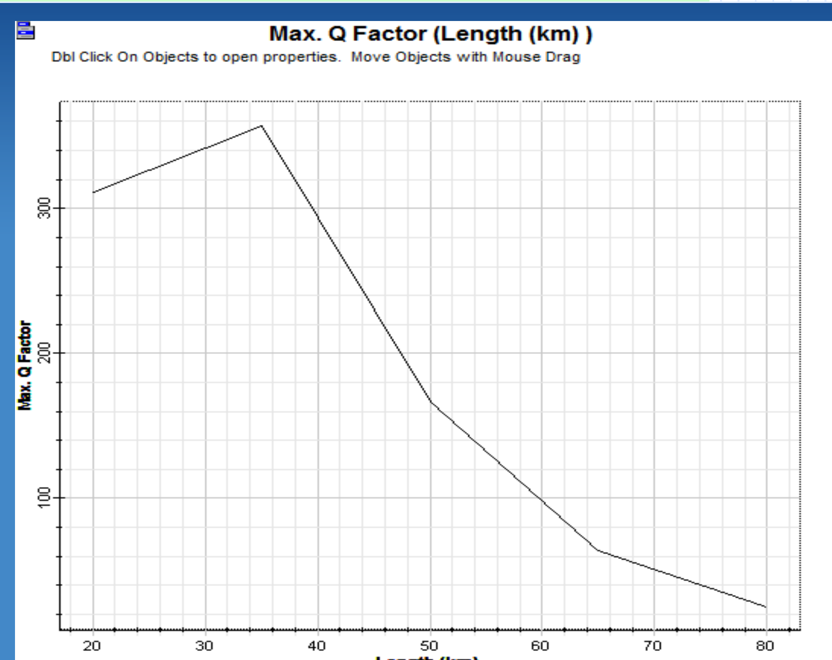
Downstream






Central Office (CO) /
Optical Line Terminal (OLT)
Feeder Fiber
Remote Nodes (RN)
Upstream Transmitter
Frequency = 1490 nm
Power = 3 dBm
Bit rate = Bit rate Bits/s
Modulation type = NRZ
Downstream Pump
Frequency = 1239 nm
Power = 980 mW
Upstream Pump
Frequency = 1427 nm
Power = 370 mW
Ideal Mux
Bidirectional Optical Fiber
Length = 80 km
OD
Delay = 1
Butterworth Optical Filter_2
Frequency = 1490 nm
Bandwidth = 5 nm
Insertion loss = 2 dB
1:64 Splitter GPON
Buffer Selector
Selection = Iterations - 1
Iterations = Iterations
Butterworth Optical Filter
Frequency = 1310 nm
Bandwidth = 5 nm
Insertion loss = 2 dB
Photodetector
LPF
Cutoff frequency = 0.75 * Bit rate Hz
3R Regenerator_1
BER Downstream
Optical Network Unit (ONU)
Photodetector PIN
Low Pass Bessel Filter
3R Regenerator
Cutoff frequency = 0.75 * Bit rate Hz
BER Upstream
OD
Delay = 1
Downstream Transmitter
Frequency = 1310 nm
Power = 0 dBm
Bit rate = Bit rate Bits/s
Modulation type = NRZ
Case II: With Raman Amplifier


Upstream
Dbl Click On Obj
with Mouse Drag
Signal Index: 0
Auto Set
Show Eye Diagram
Analysis
Max. Q Factor 25.063
Min. BER 0
Eye Height 2.15421e-005
Threshold 1.15371e-005
Decision Inst. 0.5
Invert Colors
Color Grade
Patterns
Calculate Patterns
Patterns
Pattern 1 1e-012
Pattern 2 1e-011
Pattern 3 1e-010
Pattern 4 1e-009
Pattern 5 1e-008
Q
Amplitude (a.u.)
Time (bit period)
0
0.5
1
10
20


Max. Q Factor (Length (km))
Dbl Click On Objects to open properties. Move Objects with Mouse Drag
Max. Q Factor
Length (km)
100
200
300
20
30
40
50
60
70
80

Your Exercise today

- Make the 2-km PON work
- Extend to a 20-km PON
- If you feel energetic, think about the design of the following PON

20km GPON